2.3 Solving Trigonometric Equations

Name _____ Period ____ Date _____

Pre-Calculus 2.3 (30-minute plan) · Student sheet

I can…

- solve a trig equation and give every solution, either the general solution or the solutions in [0, 2π).

Vocabulary · fill in the missing word

General solution — every solution: add 2nπ for sin and cos (period 2π) or nπ for tan (period π), where n is any _____

Solutions in [0, 2π) — only the solutions from one trip _____ the unit circle, $0 \le x < 2\pi$

Equation of quadratic type — like $2\sin^2 x - \sin x - 1 = 0$: factor it as if sin x were a _____

Multiple angle — an angle like 2x or 3t: solve for the whole angle first, then _____

Extraneous solution — a value that appears after _____ both sides but doesn't check in the original equation

Part 1 · Notes and examples (fill in each note, then work the example)

Strategies 1 and 2: 1. Isolate the _____. 2. Extracting square roots: remember the _____.

General solution: General solution: add _____ for sine and cosine, and add _____ for tangent.

Example 1

Solve: $2\sin x - 1 = 0$

Example 2

Solve: $3\tan^2 x - 1 = 0$

Strategy 3: 3. Factor. Never _____ both sides by a trig function: you lose solutions.

Strategies 4 and 5: 4. Quadratic type: factor as if sin x were a _____. 5. If you square both sides, _____ every answer.

Example 3

Solve on [0, 2π): $2\sin^2 x - \sin x - 1 = 0$

Part 2 · Partner practice

Partner practice

Solve: $\sqrt{3}\csc x - 2 = 0$

Part 3 · On your own: choose 2 of 3

Problem 1

Solve on $[0, 2\pi)$: $2\cos x + 1 = 0$

Problem 2

Solve on $[0, 2\pi)$: $\tan^2 x - 3 = 0$

Problem 3

Solve on $[0, 2\pi)$: $2\cos^2 x + \cos x - 1 = 0$